

Enhancement of Human 3D Pose Estimation Using a Novel Concept of Depth Prediction with Pose Alignment from a Single 2D Image

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Human Action Recognition

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Introduction

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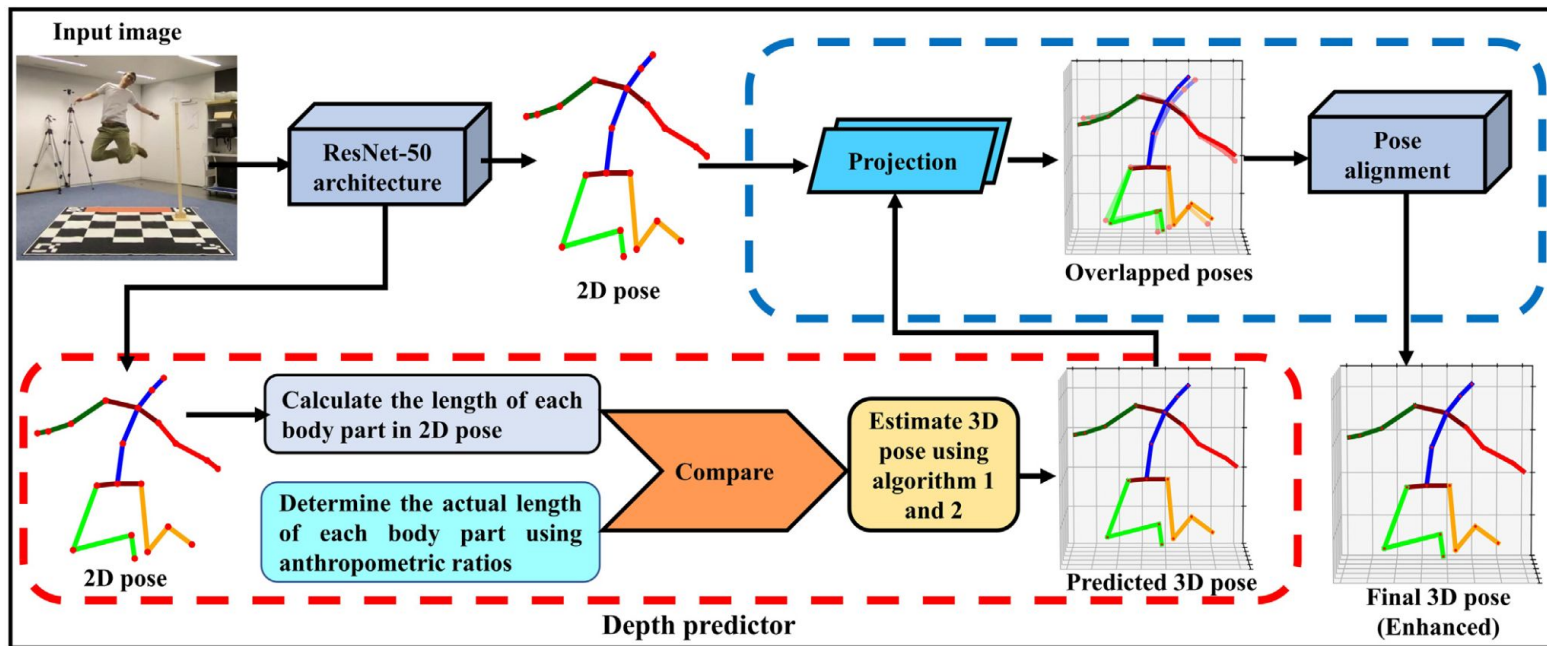
- **2D poses limited** in estimating human activities under geometric ambiguities, occlusion, diverse appearances.
- Human skeletal **3D posture is crucial** for applications like avatar creation, action recognition, AR, etc.
- Research goal is to **estimate accurate 3D human poses from a single 2D image** with minimum errors.
- **Challenges: estimating depth from a single 2D image is complex.**

Proposed methodology

1. Framework
2. Modules:
 - a. 2D Keypoints detection
 - b. Depth prediction
 - c. Pose alignment

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Framework

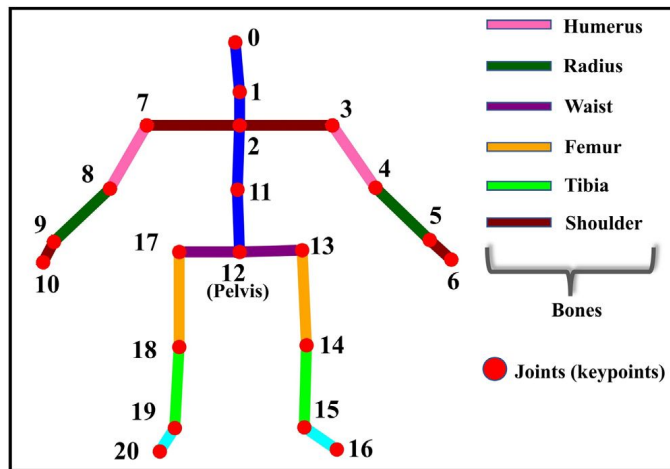


Module 1: 2D Keypoints detection

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Uses ResNet-50 to localizes 2D keypoints of the human body to become a skeleton from an RGB image.

- This model can **estimate extra** and important **keypoints of legs** in the human body.
- Provides the **foundation for 3D pose estimation**.



Module 2: Depth predictor

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Utilizes anthropometric measurements to transform a collection of 2D joint coordinates (keypoints) into 3D space.

Process:

- Algorithm 1: Calculate the angle of deviation of body parts.
- Algorithm 2: Transform 2D coordinates into 3D space.

Module 2: Depth predictor - Algorithm 1

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1. **Initialization:**

- Start with 2D coordinates (X, Y) and initialize Z to zero for 21 joints.
- Define body part pairs (B) that connect key joints.

2. **Calculate Distances:**

- For each pair in B, calculate the Euclidean distance in 2D.

3. **Anthropometric Calculations:**

- Use provided equations to relate body part lengths to height.

4. **Angle Calculation:**

- Compute the deviation angle (θ) using the ratio of measured distances to expected distances.
- Use arccosine to find the angle in radians.

Module 2: Depth predictor - Algorithm 2

1. Initialization:

- Fix the Z-coordinate of the pelvis joint to zero.
- Assume zero depth for backbone joints.

2. Construct Backbone:

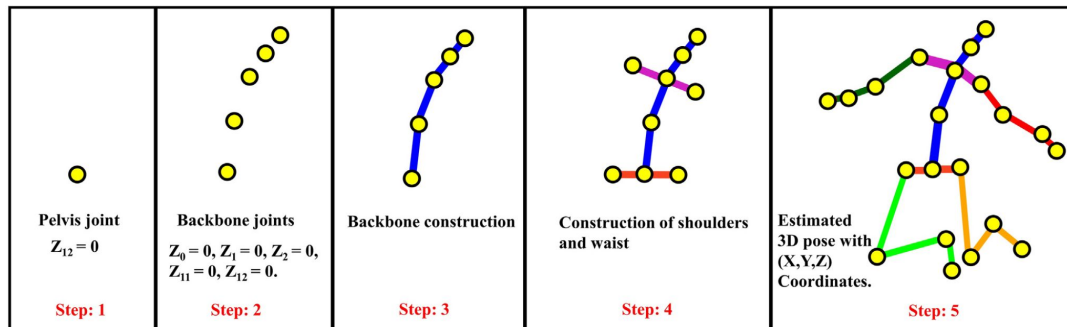
- Join the backbone keypoints.

3. Construct Shoulders and Waist:

- Use trigonometric equations to calculate positions.

4. Complete the Pose:

- Loop through other body parts and compute their Z-values.



Module 3: Pose alignment

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The objective is to reduce positional errors in the predicted 3D pose.

Process:

- Project 3D pose back to 2D plane.
- Match 2D-3D keypoints using Euclidean distance of all keypoint pairs in the image.

$$k_{2d} = s\Pi RM(\theta)k_{3d} + t$$

$$E(s, t, \theta) = \sum_{i=0}^{N-1} \|k_{2di} - \tilde{k}_{2di}\|_2^2$$

$$E(s, \theta, t) = \sum_{i=0}^{N-1} \|s\Pi RM(\theta)k_{3di} + t - \tilde{k}_{2di}\|_2^2$$

Finally, the optimal human pose is then computed using the Jacobian matrix and the Levenberg-Marquardt method.

Experimental analysis and results

1. Datasets
2. Evaluation metric
3. Comparative analysis
4. Experimental analysis
5. Strengths and limitations

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Datasets

Table 1

Overview and purpose of datasets used for the sake of experiment and results.

S. No.	Dataset used	Type (image/video)	Activity classes	Ground-truth annotation	Description
1	Human3.6M	Both	15	Yes	20 images from each activity class are used for the experiment. The MPJPE evaluation metric is used to test the accuracy of the model.
2	Human Eva-I and II	Video	6	Yes	Extract 50 images from videos of walking, jogging, and boxing activity classes. 3D pose error is used to test the accuracy of the model.
3	NTU RGB + D	Both, skeleton	60	Yes	20 images are used for experimental analysis and check the accuracy with given 3D pose annotations.
4	Articulated Free Fall	Video	7	Yes	This dataset is used for analyzing the relationship between the height and other human body parts.
5	MPII	Both	410	Yes	This dataset is used to check the effectiveness of the model under different conditions.
6	UAV-human	Both	155	Yes	This dataset is used to show the outcomes of the proposed model under the challenging scenario, like different view angles.

Evaluation metric

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Mean Per Joint Position Error (MPJPE) has been used to evaluate the positional errors in the reconstructed 3D pose.

$$MPJPE = \frac{1}{N} \sum_{i=0}^N (K^i - G^i)^2$$

MPJPE is the ‘mean of the Euclidean distance between the predicted 3D coordinates K and the Ground-truth G ’.

Comparative analysis

Table 2

For each activity category of the **Human3.6M** dataset, the table compares MPJPE values of the state-of-the-art methods with the proposed approach.

Methods	Year	Direction	Discuss	Eat	Walk	Pose	Purchase	SitDown	Phone
Ionescu [45]	2014	132.71	183.55	132.37	96.60	150.61	171.31	243.03	162.12
Tekin [51]	2016	102.41	147.72	88.83	55.07	112.38	129.17	224.9	118.02
Du [52]	2016	85.07	112.68	104.9	99.26	105.93	166.16	226.94	139.08
Chen [17]	2017	89.87	97.57	89.56	87.02	93.56	136.09	240.12	107.31
Bruce [53]	2017	90.12	88.24	85.76	90.60	90.41	117.96	98.51	103.90
Tekin [54]	2017	53.91	62.19	61.51	52.81	64.61	83.17	107.92	80.12
Zhou [4]	2017	54.82	60.70	58.22	63.22	53.83	55.58	111.59	62.03
LCR_Net++ [55]	2019	55.98	65.32	68.14	47.83	71.81	62.56	94.65	69.89
Wang [56]	2019	50.03	59.96	54.66	41.43	54.32	54.81	117.98	65.65
Xu [57]	2020	37.40	43.50	42.70	33.70	41.30	45.10	60.20	46.60
Wang [58]	2020	38.20	41.00	45.90	33.10	41.60	41.40	57.40	41.40
Chen [59]	2021	41.40	43.50	40.10	31.50	41.70	42.30	60.20	41.70
Li [60]	2022	40.30	43.30	40.20	30.00	41.80	40.50	60.60	45.60
Proposed method		38.58	40.05	39.56	32.46	40.49	40.67	51.14	40.51
Methods	Year	Smoke	Photo	Wait	WalkDog	Sit	WalkPair	Greet	Average
Ionescu [45]	2014	162.14	205.94	170.69	177.13	151.57	127.88	164.39	162.14
Tekin [51]	2016	118.42	182.64	138.75	126.29	138.89	65.76	125.28	124.97
Du [52]	2016	120.02	135.91	117.56	137.36	117.49	106.54	122.05	126.46
Chen [17]	2017	106.65	139.18	106.21	114.05	133.14	90.55	107.38	115.88
Bruce [53]	2017	103.01	92.42	94.40	86.05	136.40	89.51	95.60	97.53
Tekin [54]	2017	70.44	79.45	68.01	77.81	70.93	63.11	66.18	70.81
Zhou [4]	2017	64.15	65.53	66.05	51.43	75.20	55.33	71.41	64.61
LCR_Net++ [55]	2019	90.76	67.93	52.69	63.31	55.32	54.61	56.36	65.14
Wang [56]	2019	62.48	79.63	59.55	65.21	85.85	48.52	56.55	63.78
Xu [57]	2020	45.80	59.70	43.10	47.70	52.70	37.10	42.70	45.60
Wang [58]	2020	41.80	51.40	44.40	41.60	52.00	30.00	39.70	42.60
Chen [59]	2021	45.40	51.90	41.70	46.00	53.90	32.70	42.90	44.10
Li [60]	2022	44.20	52.30	43.00	44.20	55.90	30.20	42.30	43.70
Proposed method		41.25	44.21	41.26	40.10	48.84	31.56	38.40	40.61

Comparative analysis

Table 3

Comparative analyses of the proposed model with the state-of-the-art methods on **HumanEva-I** dataset. “-” in the table indicates that authors of the prior research does not compare their results on that activity class of the dataset.

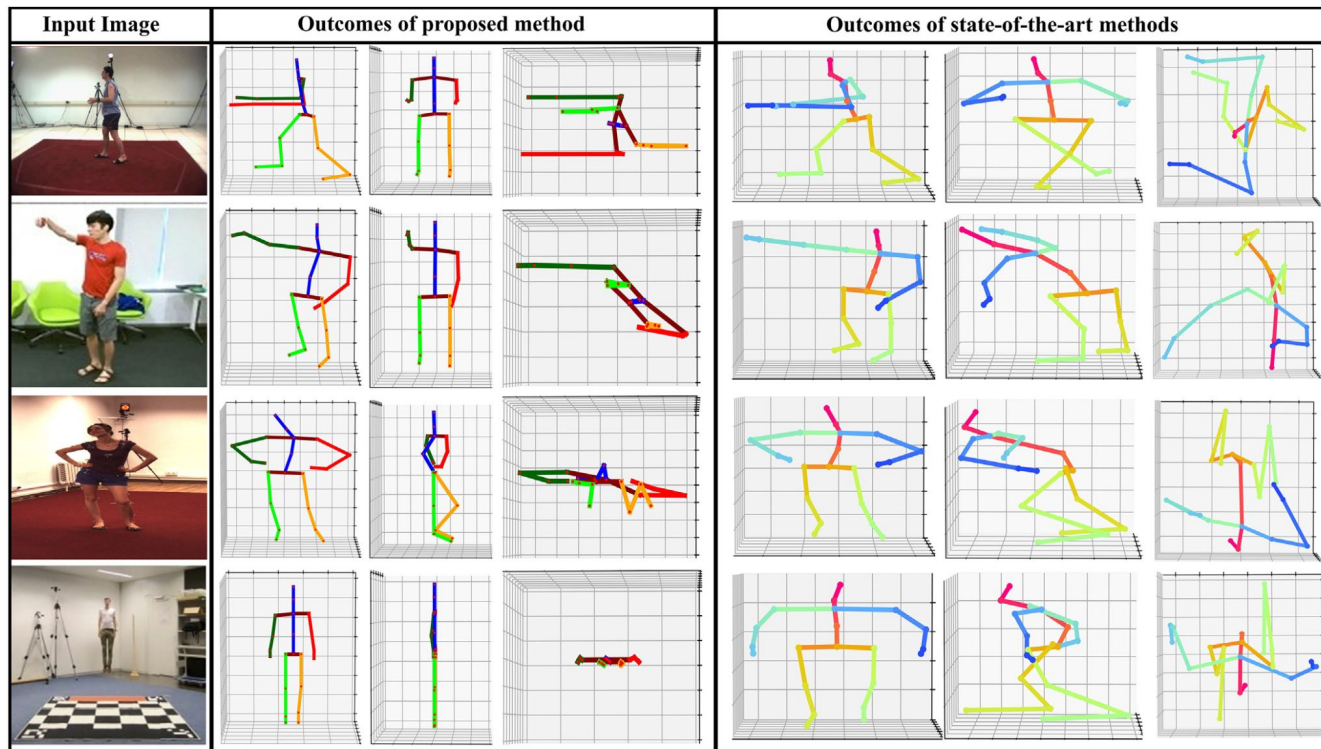
Methods	Year	Walking				Jogging				Boxing			
		S1	S2	S3	Avg.	S1	S2	S3	Avg.	S1	S2	S3	Avg.
Du [52]	2016	62.2	61.9	69.2	64.4	56.3	59.3	59.3	58.3	-	-	-	-
Tekin [51]	2016	37.5	25.1	49.2	37.3	-	-	-	-	50.5	61.7	57.5	56.6
Lin [61]	2017	26.5	20.7	38.0	28.4	41.0	29.7	29.1	33.2	39.4	57.8	61.2	52.8
Pavlakos [2]	2017	22.3	19.5	29.7	23.8	28.9	21.9	23.8	24.9	-	-	-	-
Tekin [54]	2017	27.2	14.3	31.7	24.4	-	-	-	-	-	-	-	-
Cheng [62]	2019	11.7	10.1	22.8	14.8	18.7	11.4	11.0	13.7	-	-	-	-
Cheng [63]	2020	10.6	11.8	19.3	13.9	15.8	11.5	12.2	13.1	-	-	-	-
Li [60]	2022	14.0	10.0	32.8	18.9	19.5	13.6	14.2	15.7	22.4	21.6	22.5	22.1
Proposed method		11.4	10.2	20.4	14.0	15.0	10.6	12.1	12.6	20.2	21.0	20.4	20.5

Table 4

Experimental analysis of the proposed model (depth predictor with or without pose alignment) Under different settings of the 2D pose estimation architectures on different datasets.

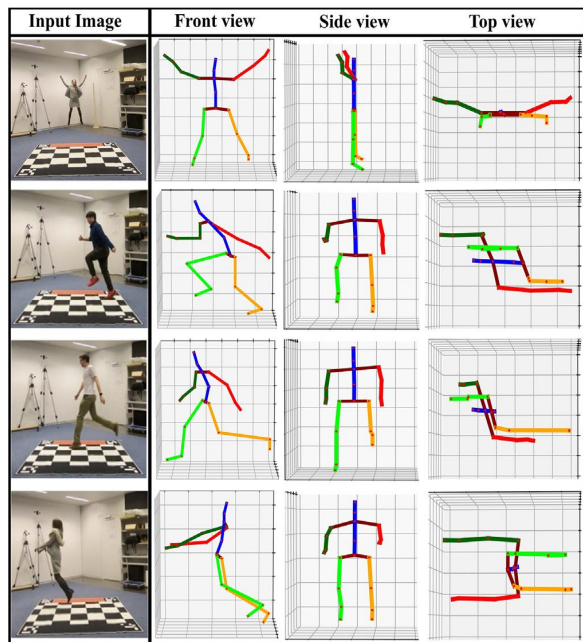
Methods	Human3.6M	Human Eva-I	NTU RGB + D	UAV-human
OpenPose [16] w/o pose alignment	65.29	42.60	62.54	65.42
AlphaPose [29] w/o pose alignment	62.40	40.20	68.49	63.45
ResNet-50 [15] w/o pose alignment	48.30	24.30	57.5	59.88
ResNet-50 with pose alignment	40.61	15.70	45.8	48.65

Comparative analysis

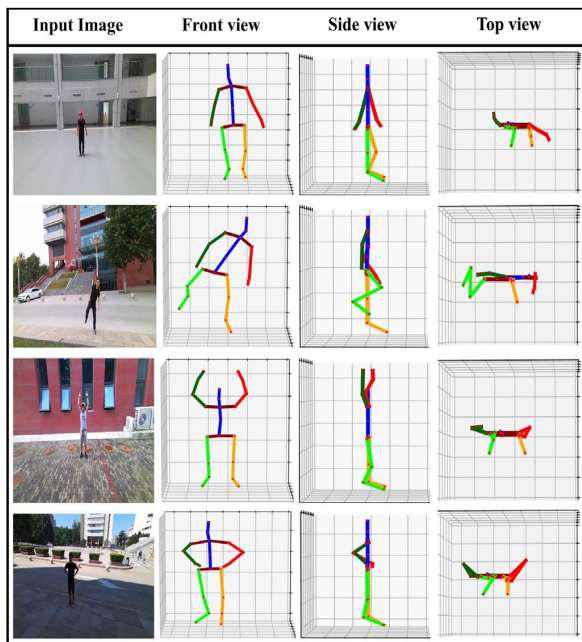


Experimental analysis

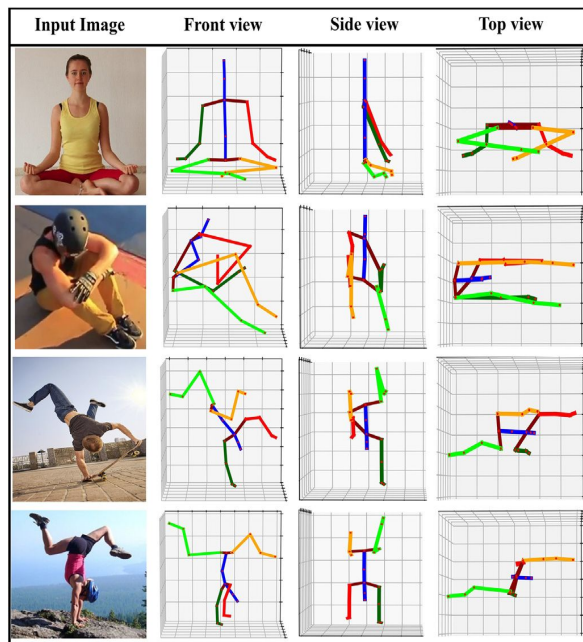
Articulated Free Fall Dataset



UAV-Human Dataset

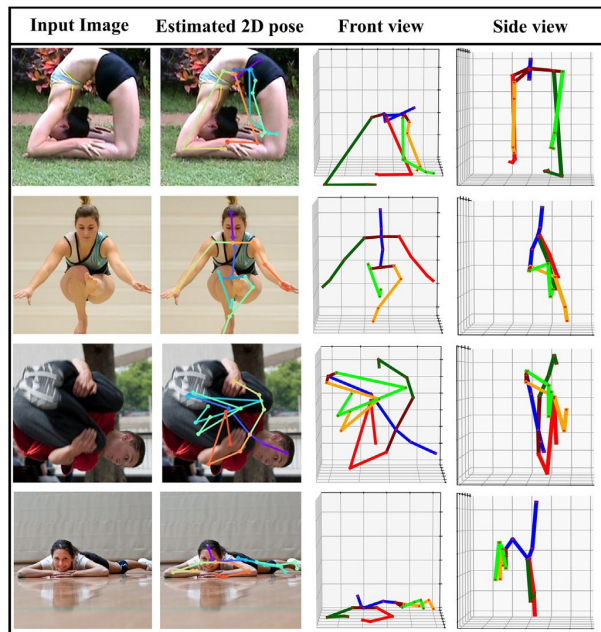


MPII Dataset

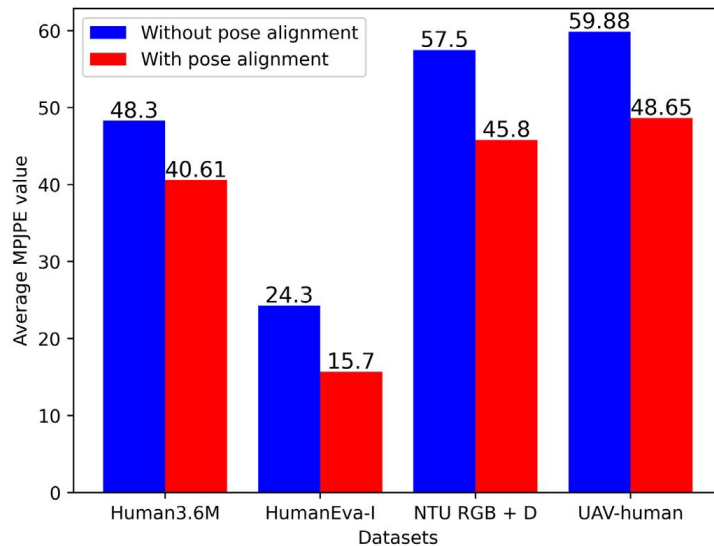


Experimental analysis

Some failure cases: due to inaccurate prediction of 2D pose on MPII dataset.



Effectiveness of depth prediction with or without pose alignment module on different datasets.



Conclusion

- The novel approach **effectively estimates 3D human poses** from 2D images with reduced errors
- Outperforms the state-of-the-art methods in terms of positional error and computational time.
- **No training dataset** required.
- **Effective** under various **scale variations**.
- Limitations:
 - For a more accurate 3D pose, a **2D posture must be precise**.
 - The method **utilizes** the relationship between the human height and different **body parts of standard human** body structure.
- Future work: explore further improvements and real-world applications.

Thank you!